Impulse changes momentum

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect force over time to momentum change.

SUCCESS CRITERIA - I can...

1. Calculate impulse from force and duration.
2. Match impulse direction to momentum change.
3. Use impulse to find a final velocity.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A planet has surface field 75 N/kg. At center distance 5 R, find field and the force on a 7 kg object.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $J = F_{avg}\,\Delta t = \Delta p = p_f - p_i$.
J is impulse; F_{avg} is average net force. Signed force-time area gives impulse; i = initial, f = final.

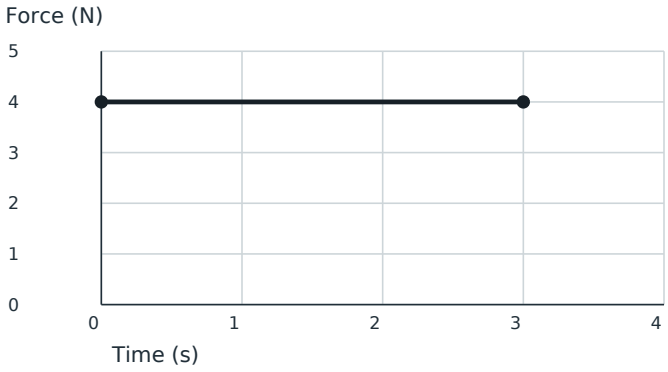
Worked reasoning: A constant 6 N force acts right for 2 s. Impulse is 12 N·s right, so momentum increases by 12 kg·m/s.

Your turn: Draw the force-time rectangle and shade its area. Explain why the two impulse units are equivalent.

Marzano check - circle: ① ② ③

03 Look and explain

Read the rectangular force-time graph. Find the area and state what it means.



Marzano check - circle: ① ② ③

04 Build the solution

A 2 kg cart starts at rest and receives 6 N·s rightward impulse. Final momentum is 6 kg·m/s. Find final velocity by dividing by mass.

Marzano check - circle: ① ② ③

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05 Use the idea

Two force pulses have equal signed area but different peak heights. What must be equal? What need not be equal?

Marzano check - circle: 1 2 3

06 Choose a method

A spherical planet has surface field 144 N/kg. Find the field at center distance 6 R, where R is the planet radius.

Marzano check - circle: 1 2 3

07 Exit check

A 5 N force acts right for 2 s. Find impulse. On a 2 kg cart initially at rest, find final velocity.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Create two constant force-and-time pairs giving an impulse of 12 N·s. Compare their force-time rectangles.

Marzano check - circle: 1 2 3